

# Speaker Notes

## AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media

Diploma defence speech notes

Bachelor's diploma defence, 2026

### Contents

|                                                                                     |    |
|-------------------------------------------------------------------------------------|----|
| Slide 1. AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media | 3  |
| Slide 2. Relevance of the Topic                                                     | 3  |
| Slide 3. Research Problem                                                           | 4  |
| Slide 4. Aim and Tasks                                                              | 4  |
| Slide 5. Object and Subject of the Study                                            | 4  |
| Slide 6. Thermoelastic Coupling Logic                                               | 5  |
| Slide 7. Geological Media and Effective Parameters                                  | 5  |
| Slide 8. Mechanical Rock Parameters                                                 | 6  |
| Slide 9. Thermal and Porosity Parameters                                            | 6  |
| Slide 10. Rock Properties Controlling Propagation                                   | 7  |
| Slide 11. Elastic Waves in Geological Media                                         | 7  |
| Slide 12. Heat Conduction and Thermal Expansion                                     | 8  |
| Slide 13. Key Physical Parameters                                                   | 8  |
| Slide 14. Thermoelastic Coupling                                                    | 8  |
| Slide 15. Mathematical Scope and Primary Fields                                     | 9  |
| Slide 16. Directional Prediction Task                                               | 9  |
| Slide 17. COMSOL-Based Synthetic Dataset                                            | 10 |

|                                                  |    |
|--------------------------------------------------|----|
| Slide 18. Data Representation and Preprocessing  | 10 |
| Slide 19. Comparative Prototype Framework        | 11 |
| Slide 20. Shared Model Comparison Contract       | 11 |
| Slide 21. Physics-Informed Neural Network (PINN) | 12 |
| Slide 22. Fourier Neural Operator (FNO)          | 12 |
| Slide 23. MeshGraphNet                           | 13 |
| Slide 24. Transformer-Style Surrogate Model      | 13 |
| Slide 25. Calculation Status                     | 13 |
| Slide 26. Travel-Time Predictions                | 14 |
| Slide 27. Accuracy Against Analytical Reference  | 14 |
| Slide 28. Inference Latency and Time Behavior    | 15 |
| Slide 29. Threats to Validity                    | 15 |
| Slide 30. Conclusions                            | 16 |
| Slide 31. Future Work                            | 16 |
| Slide 32. Thank You                              | 17 |
| Possible questions from the committee            | 18 |
| Short version of the speech                      | 19 |

## Slide 1. AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media

### Purpose:

Open the defence, name the topic, and immediately set the correct research scope.

### What to say:

Good morning. The topic of our diploma work is *AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media*. The work studies how machine learning can support directional prediction for thermoelastic wave behavior in simplified geological media. I would like to emphasize the scope from the beginning: this is a research prototype and a comparative framework. It is not presented as a site-calibrated simulator or as a replacement for numerical methods. The focus is on physical interpretation, model comparison, and estimating directional characteristics under simplified assumptions.

### Key points:

- The work connects thermoelastic physics, geological parameters, and neural surrogate models.
- The main target is directional prediction between a source and a probe point.
- The presentation treats the system as a research prototype.

### Transition to next slide:

I will start by explaining why this topic is relevant.

---

## Slide 2. Relevance of the Topic

### Purpose:

Explain why thermoelastic wave prediction in rocks is worth studying.

### What to say:

Thermally affected rocks appear in several applied contexts: geothermal energy, underground construction, thermal stimulation, and geophysical interpretation. When temperature changes inside a rock, it can influence stress, deformation, and wave behavior. A direct coupled numerical calculation is useful, but it can be expensive when many materials, source positions, or observation directions must be compared. This motivates an artificial intelligence-assisted approach, where neural models estimate selected directional responses and help compare scenarios faster. The thesis is positioned as a research prototype: it compares surrogate approaches under simplified assumptions rather than replacing numerical solvers.

### Key points:

- Temperature changes can create mechanical effects in geological media.
- Repeated coupled simulations are expensive for comparative studies.
- The work uses AI-assisted prediction as a supporting research tool.

### Transition to next slide:

This relevance leads to the main research problem.

---

## Slide 3. Research Problem

### **Purpose:**

State the central question and narrow it to a defensible bachelor thesis scope.

### **What to say:**

The research problem is to estimate directional characteristics of thermoelastic wave propagation in geological media using an AI-assisted comparative framework under simplified physical assumptions. This is important because coupled thermal and mechanical calculations can become costly when many materials and directions are compared. The work estimates selected source–probe responses and travel-time characteristics in a controlled model-based calculation setting. It does not claim a complete field-scale simulator or a field-validated prediction system.

### **Key points:**

- The problem is framed as directional prediction, not full field simulation.
- The work is important for comparative studies with many query directions.
- The simplified setup makes controlled model comparison possible.

### **Transition to next slide:**

Next, I will formulate the aim and the tasks of the work.

---

## Slide 4. Aim and Tasks

### **Purpose:**

Show what the diploma work tries to achieve and how the study is organized.

### **What to say:**

The aim is to develop and justify an artificial intelligence-assisted framework for directional prediction of thermoelastic wave propagation in geological media. The calculations were performed for representative geological materials, including granite, basalt, sandstone, and limestone. To reach the aim, the work establishes the physical basis, formulates a simplified two-dimensional prediction task, prepares COMSOL-derived synthetic data, compares four surrogate routes, and interprets the results and limitations.

### **Key points:**

- The aim combines physical reasoning with model comparison.
- The task is built around a simplified two-dimensional formulation.
- The representative materials are granite, basalt, sandstone, and limestone.

### **Transition to next slide:**

Before the theory, I will define the object and subject of the study.

---

## Slide 5. Object and Subject of the Study

### **Purpose:**

Clarify what is being studied and what specific aspect the thesis focuses on.

**What to say:**

The object of the study is thermoelastic wave propagation in geological media represented by effective rock parameters. The subject is narrower: artificial intelligence-assisted directional prediction of thermal and displacement response along a selected source–probe path. This distinction is important. We do not model a full natural formation with all geological details. We study a simplified comparative framework where the material is represented by effective parameters and the prediction is made for a defined direction.

**Key points:**

- Object: thermoelastic wave behavior in geological media.
- Subject: directional prediction along a source–probe path.
- The work is intentionally limited to a simplified comparative framework.

**Transition to next slide:**

Now I will explain the basic physical logic of thermoelastic coupling.

---

## Slide 6. Thermoelastic Coupling Logic

**Purpose:**

Introduce the physical chain from heating to wave-like mechanical response.

**What to say:**

This slide shows the basic logic of thermoelastic coupling. A local thermal source changes the temperature field. Temperature change produces thermal strain, and if the expansion is constrained by the surrounding material, stress appears. Stress gradients then create mechanical displacement, and this response propagates through the medium. The chain at the bottom is a physical summary rather than a full equation:  $T$  is temperature in kelvin,  $\varepsilon_{\text{th}}$  is dimensionless thermal strain,  $\sigma$  is stress in pascals, and  $u$  is displacement in metres.

**Key points:**

- Heating changes the temperature field.
- Thermal expansion can produce stress.
- Stress gradients drive displacement and propagation.

**Transition to next slide:**

The next group of slides explains the physical and geological parameters used in this chain.

---

## Slide 7. Geological Media and Effective Parameters

**Purpose:**

Show that geological materials enter the prototype through effective scalar parameters.

**What to say:**

Here we show a compact example of the geological material parameters used in the study. Density, compressional wave velocity, thermal conductivity, and heat capacity differ between rock types such as granite, basalt, sandstone, and limestone. In the prototype, these are effective scalar parameters.

They allow us to compare material behavior in a controlled way. The limestone texture is included only as an illustrative geological material image; it is not used as input data.

**Key points:**

- Rock types differ in density, stiffness, wave speed, and thermal properties.
- The parameters are effective values for comparison.
- The table supports modelling assumptions rather than field calibration.

**Transition to next slide:**

The next slide separates the mechanical part of the material catalogue.

---

## Slide 8. Mechanical Rock Parameters

**Purpose:**

Document the mechanical part of the material catalogue used for comparison.

**What to say:**

This table separates the mechanical parameters from the larger material catalogue. It lists density, Young's modulus, Poisson's ratio, shear modulus, bulk modulus, and representative compressional and shear wave velocities. I will not read every value. The purpose is to show that the model inputs are physically meaningful and material-dependent, while still being effective midpoint values used under isotropic assumptions.

**Key points:**

- Mechanical parameters control inertia, stiffness, and elastic wave speeds.
- The values are effective and comparative, not site-calibrated.
- Vertical table boundaries separate values and units for readability.

**Transition to next slide:**

The next slide gives the thermal and porosity part of the catalogue.

---

## Slide 9. Thermal and Porosity Parameters

**Purpose:**

Document the thermal and porosity parameters without overloading one table.

**What to say:**

This table gives thermal expansion, thermal conductivity, heat capacity, volumetric heat capacity, thermoelastic coupling coefficient, and porosity information. These quantities control how a temperature disturbance spreads and how temperature change contributes to thermal strain and stress. Missing values are shown as dashes rather than filled by unsupported assumptions. This keeps the material representation transparent.

**Key points:**

- Thermal parameters control heat spreading and thermal strain.
- Porosity is included as an effective descriptive property.

- Missing values are explicit and are not silently invented.

**Transition to next slide:**

After listing the parameters, I will explain how they control propagation.

---

## Slide 10. Rock Properties Controlling Propagation

**Purpose:**

Explain the physical role of mechanical and thermal rock properties.

**What to say:**

Rock properties influence propagation through both mechanical and thermal mechanisms. Density controls inertia: a denser medium resists acceleration more strongly. Elastic stiffness controls how strongly the rock resists deformation. Porosity, cracks, and saturation can reduce effective stiffness and change wave behavior. On the thermal side, conductivity controls heat transfer, heat capacity controls thermal storage, and thermal expansion controls how temperature change becomes strain. In real rocks, bedding, foliation, aligned pores, fractures, and stress state can also create directional effects.

**Key points:**

- Mechanical properties control inertia and stiffness.
- Thermal properties control heat spreading and thermal strain.
- Geological structure can create directional behavior.

**Transition to next slide:**

To connect this with wave propagation, I will briefly discuss P-waves and S-waves.

---

## Slide 11. Elastic Waves in Geological Media

**Purpose:**

Introduce compressional and shear waves using simple physical interpretation.

**What to say:**

Elastic wave propagation is commonly described using compressional waves and shear waves. Compressional waves have particle motion mainly parallel to the propagation direction, while shear waves have transverse motion and depend on shear rigidity. The formulas define  $V_P$  and  $V_S$  in metres per second. They depend on bulk modulus  $K$ , shear modulus  $G$ , and density  $\rho$ . In physical terms, larger stiffness tends to increase wave velocity, while larger density increases inertia.

**Key points:**

- P-waves involve compression and volume change.
- S-waves depend on shear rigidity.
- Wave velocities depend on elastic moduli and density.

**Transition to next slide:**

Next, I will add the thermal part of the coupled problem.

---

## Slide 12. Heat Conduction and Thermal Expansion

### Purpose:

Explain how heat transfer enters the thermoelastic formulation.

### What to say:

The thermal part is described by Fourier heat flux and the heat equation. In the notation,  $q$  is heat flux in watts per square metre,  $k$  is thermal conductivity,  $T$  is temperature in kelvin,  $\rho$  is density,  $C_p$  is heat capacity, and  $Q$  is a volumetric heat source. The second relation defines temperature perturbation and thermal strain:  $\theta$  and  $\Delta T$  are temperature changes in kelvin,  $\alpha$  is thermal expansion, and  $\varepsilon_{th}$  is dimensionless thermal strain. The physical meaning is simple: heat diffuses through the rock, and temperature increase can create strain.

### Key points:

- Thermal conductivity controls the heat flux.
- The heat equation describes temperature evolution.
- Thermal expansion links temperature change to strain.

### Transition to next slide:

Now I will summarize the key physical parameters in one place.

---

## Slide 13. Key Physical Parameters

### Purpose:

Make the parameter roles easy to remember for the committee.

### What to say:

This slide summarizes thermal diffusivity. The relation  $\alpha_T = k/(\rho C_p)$  combines thermal conductivity, density, and heat capacity. The result has units of square metres per second and describes how quickly a temperature disturbance spreads through the medium. This parameter is useful because it turns several material properties into one interpretable quantity related to thermal spreading.

### Key points:

- Thermal diffusivity describes the rate of heat spreading.
- Density and stiffness influence mechanical wave behavior.
- Thermal expansion is the bridge from heat to deformation.

### Transition to next slide:

The next slide combines these ideas in the thermoelastic equations.

---

## Slide 14. Thermoelastic Coupling

### Purpose:

Present the theoretical backbone without overloading the speech with derivation.

### What to say:



This slide gives the thermoelastic extension of Hooke's law. The first relation defines temperature perturbation relative to the reference temperature. The stress relation contains the elastic part, controlled by the Lamé parameters, and the thermal part, controlled by the coupling coefficient  $\gamma = 3K\alpha$ . Stress is measured in pascals, strain is dimensionless, and displacement is measured in metres. The equation of motion then links stress gradients to acceleration and displacement. In simple terms, temperature changes can modify stress, and stress drives motion.

**Key points:**

- Stress contains both elastic and thermal contributions.
- The thermal coupling coefficient is related to bulk modulus and expansion.
- The equation of motion connects stress gradients with displacement.

**Transition to next slide:**

I will now define the mathematical scope used in the thesis.

---

## Slide 15. Mathematical Scope and Primary Fields

**Purpose:**

Make the simplifications and primary fields explicit and defensible.

**What to say:**

The model is formulated as a two-dimensional continuum problem. The domain  $\Omega$  is the model region, position is measured in metres, time is measured in seconds, and the displacement components are measured in metres. Small deformation and linear isotropic thermoelasticity are used as first-order approximations. The table also lists the primary fields: temperature, temperature perturbation, displacement, strain, stress, and direction variables. The model does not explicitly resolve pores, fractures, fluids, plasticity, damage, or full anisotropic constitutive behavior.

**Key points:**

- The thesis uses a two-dimensional continuum formulation.
- Small deformation and linear isotropic thermoelasticity are assumed.
- Complex geological effects are acknowledged as limitations.

**Transition to next slide:**

With the assumptions defined, I will explain the directional prediction task.

---

## Slide 16. Directional Prediction Task

**Purpose:**

Define the source-probe query and what the prototype estimates.

**What to say:**

Directional prediction is built around a source point and a probe point. The vector from the source to the probe defines the direction of interest. Here  $x_s$  and  $x_p$  are positions in metres,  $d$  is the direction vector,  $L$  is distance,  $\hat{d}$  is a dimensionless unit direction, and  $\phi$  is the azimuth angle in radians. The mapping  $\mathcal{F} : X \rightarrow \hat{Y}$  means that the model maps input features to estimated outputs. We are

not predicting every possible field value in a complete geological volume; we estimate directional characteristics for a specified query.

**Key points:**

- A source and a probe define the directional query.
- Distance and azimuth are explicit geometric quantities.
- The predicted response is interpreted along that direction.

**Transition to next slide:**

To train and compare the models, the work uses a controlled synthetic dataset.

---

## Slide 17. COMSOL-Based Synthetic Dataset

**Purpose:**

Explain where the controlled training and comparison data come from.

**What to say:**

The dataset is derived from a controlled COMSOL-based thermoelastic calculation. The setup is a two-dimensional plane-strain rock domain with a heated internal source, used to generate time-dependent synthetic data. I will not focus on the initial conditions in the speech. The initial and boundary assumptions are used only to define a consistent calculation setting and are not the main focus of the presentation. The purpose of the dataset is to provide a common reference for comparing surrogate models under the same conditions.

**Key points:**

- The dataset comes from a controlled COMSOL-based calculation.
- The model uses a two-dimensional plane-strain domain.
- The data provide a shared basis for comparison.

**Transition to next slide:**

After data generation, the exports must be represented in a model-ready form.

---

## Slide 18. Data Representation and Preprocessing

**Purpose:**

Show how raw simulation exports are prepared for different model families.

**What to say:**

The COMSOL comma-separated value exports are aligned into a single node set. Dynamic features are assembled into a spatio-temporal tensor  $\mathbf{S}$  with dimensions for time steps, spatial points, and feature channels. Static material and geometry features are stored separately. Before training, the feature channels are normalized using the mean and standard deviation. In the normalization formula,  $\tilde{x}$  is the normalized feature,  $x$  is the original feature in its native unit, and  $\varepsilon$  is a small stabilizer.

**Key points:**

- Raw exports are aligned by nodes and time steps.

- Dynamic and static features are represented separately.
- Normalization makes model training and comparison more stable.

**Transition to next slide:**

Now I will describe the comparative prototype framework.

---

## Slide 19. Comparative Prototype Framework

**Purpose:**

Explain that all models are evaluated under one shared workflow.

**What to say:**

The prototype uses one workflow for all model routes. The frontend, backend orchestrator, material catalog, and model services follow a shared input-output contract. This is important for fair comparison: the same source–probe geometry, the same material presets, and the same normalized response format are used for every route. The goal is not to show a production platform, but to create a consistent research environment for comparing AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media methods.

**Key points:**

- All model routes receive comparable inputs.
- Outputs are normalized into the same response structure.
- The framework supports fair model comparison.

**Transition to next slide:**

Next, I will introduce the four model families compared in the work.

---

## Slide 20. Shared Model Comparison Contract

**Purpose:**

Explain the shared input-output contract before discussing individual models.

**What to say:**

All routes are treated as surrogate modelling approaches for the same directional prediction task. The input side includes material parameters, source–probe geometry, observation time, and processed field features. The output side includes estimated temperature, displacement components, travel time, and diagnostic quantities. The comparison uses the same material presets, geometry contract, response format, and analytical travel-time sanity check. This keeps the comparison fair and avoids presenting the models as complete numerical solvers.

**Key points:**

- The comparison is based on one shared prediction contract.
- Inputs and outputs are kept consistent across all routes.
- The models are surrogate approaches, not equivalent physical solvers.

**Transition to next slide:**

I will now discuss each model route separately, starting with the physics-informed baseline.

---

## Slide 21. Physics-Informed Neural Network (PINN)

### **Purpose:**

Explain why PINN is treated differently from the purely supervised surrogate routes.

### **What to say:**

The Physics-Informed Neural Network is the physics-informed baseline. It takes coordinates, time, and material parameters as input and returns estimated temperature and in-plane displacement. The model relation shows the input variables and predicted outputs. The loss combines supervised terms with velocity, elastic, and thermal residual terms. The important interpretation is cautious: including residuals makes this route physics-informed, but it is still a baseline model inside the simplified prototype, not an exact equation solver.

### **Key points:**

- PINN maps coordinates, time, and material parameters to response fields.
- The loss includes supervised and physics-related residual terms.
- PINN is a physics-informed baseline, not a complete solver.

### **Transition to next slide:**

The next slide gives the Fourier Neural Operator route.

---

## Slide 22. Fourier Neural Operator (FNO)

### **Purpose:**

Describe the neural operator route and its limitation.

### **What to say:**

The Fourier Neural Operator route is a neural operator surrogate for structured two-dimensional fields. Its input is grid-based material and response information together with the shared source-probe and time query context. Its output is a field-style response that can be used to derive directional quantities. It is included because Fourier operator methods can capture broad field patterns. In the current prototype, however, its output scale needs recalibration, so its results are interpreted as diagnostic.

### **Key points:**

- FNO is the structured-grid neural operator surrogate.
- It is useful for broad field patterns.
- The current route has a calibration limitation.

### **Transition to next slide:**

The next model route represents the mesh as a graph.

---

## Slide 23. MeshGraphNet

### **Purpose:**

Explain the graph-based surrogate route.

### **What to say:**

MeshGraphNet is the graph-based surrogate model. Its input includes mesh nodes, graph connectivity, material features, and the query context. Its output is a node-level or probe-level response estimate after graph message passing. It is included because graph structure can represent local mesh interactions more naturally than a fixed regular grid. The current limitation is temporal: in this prototype the route behaves mainly as a fixed-horizon predictor and needs stronger time-conditioning.

### **Key points:**

- MeshGraphNet uses graph connectivity to represent local interactions.
- It is useful for mesh-aware surrogate prediction.
- Its current time behavior is limited.

### **Transition to next slide:**

The fourth route is the Transformer-style surrogate.

---

## Slide 24. Transformer-Style Surrogate Model

### **Purpose:**

Explain the attention-based surrogate route.

### **What to say:**

The Transformer-style model is an attention-based baseline surrogate for fast query prediction. Its input is a set of tokens describing geometry, material parameters, state variables, and requested observation time. Its output is a fast estimate of temperature, displacement, travel time, and diagnostics. It is included because it is flexible and low-latency. Its limitation is that it has fewer explicit physics priors, so results should be interpreted cautiously.

### **Key points:**

- The model is useful for fast flexible inference.
- It uses query-like tokens for geometry and material information.
- Its physics constraints are weaker than in the PINN route.

### **Transition to next slide:**

Now I will move from methodology to the calculation setup and results.

---

## Slide 25. Calculation Status

### **Purpose:**

Summarize the evaluation setup and make caveats transparent before showing plots.

### **What to say:**

The calculation setup uses forty scenarios per material and four materials: granite, sandstone, limestone, and basalt. With four model routes, this gives six hundred forty model responses under the same contract. Every plotted response is an active model output, not a mock fallback. The operating envelope is still limited to the two-dimensional plane-strain setup. The Fourier Neural Operator is included, but its output scale is currently miscalibrated, so it should be interpreted as a diagnostic branch rather than a reliable final route.

**Key points:**

- The comparison covers  $40 \times 4 \times 4 = 640$  model responses.
- All plotted responses are active outputs.
- FNO is included, but its current calibration is limited.

**Transition to next slide:**

The first result concerns predicted travel times.

---

## Slide 26. Travel-Time Predictions

**Purpose:**

Explain what the travel-time plot shows and what conclusion should be drawn.

**What to say:**

This plot compares predicted travel time by model and material. The main observation is that the differences between model families are larger than the difference between basalt and sandstone in the current prototype. PINN and Transformer produce much smaller travel-time estimates, while FNO and MeshGraphNet produce larger values. This does not mean one model is universally correct. It shows that, under the current setup, model formulation strongly influences directional travel-time estimates. Therefore, model comparison is necessary before drawing physical conclusions.

**Key points:**

- The plot compares travel-time estimates across models and materials.
- Model-family differences dominate the material difference here.
- The result motivates careful interpretation and validation.

**Transition to next slide:**

To judge accuracy, the next slide compares the estimates with an analytical reference.

---

## Slide 27. Accuracy Against Analytical Reference

**Purpose:**

Present the key quantitative result and its cautious interpretation.

**What to say:**

Here the predicted travel times are compared with a simple analytical reference,  $t_{\text{gt}} = r/V_P$ . In this formula,  $t_{\text{gt}}$  is reference travel time in seconds,  $r$  is source–probe distance in metres, and  $V_P$  is compressional wave velocity in metres per second. This reference is not a full thermoelastic validation; it is a useful sanity check. In this comparison, PINN has the lowest median relative error and remains within the ten percent band across the balanced sweep.

**Key points:**

- The reference is a sanity check, not complete validation.
- PINN shows the best agreement with the analytical reference.
- Other routes need further calibration or time-conditioning.

**Transition to next slide:**

Accuracy is only one side of the comparison; inference time is also important.

---

## Slide 28. Inference Latency and Time Behavior

**Purpose:**

Compare practical response time and explain the fixed-horizon issue.

**What to say:**

This plot shows inference latency by model. All routes return predictions in milliseconds on the development machine, so the prototype is suitable for interactive comparison. The Transformer route is the fastest, followed by MeshGraphNet, FNO, and PINN. However, latency is not the only criterion. PINN is also the only route that changes smoothly with requested observation time. The other routes currently behave more like fixed-horizon predictors. This means they may be fast, but their temporal behavior needs additional development for time-dependent directional studies.

**Key points:**

- All routes are fast enough for interactive prototype use.
- Transformer has the lowest median latency.
- PINN currently has the most meaningful time-conditioned behavior.

**Transition to next slide:**

Before concluding, I will state the main threats to validity.

---

## Slide 29. Threats to Validity

**Purpose:**

Show scientific caution and define the limits of the results.

**What to say:**

There are several threats to validity. First, the training data are limited and come from a pilot COMSOL-derived configuration, not from a complete calibrated geological database. Second, the model uses two-dimensional plane-strain assumptions and effective material parameters. Third, the current Fourier Neural Operator route has a normalization issue and is not yet physically reliable. Fourth, the analytical reference based on  $r/V_P$  is only a sanity check, not a full validation of thermoelastic behavior. These limitations are important because the results should be interpreted within the simplified assumptions of the study.

**Key points:**

- The dataset is limited and synthetic.
- The physical formulation is simplified to two-dimensional effective media.

- The analytical reference is useful but limited.

**Transition to next slide:**

With these limitations in mind, I can summarize the conclusions.

---

## Slide 30. Conclusions

**Purpose:**

State the main scientific outcomes of the work.

**What to say:**

The thesis develops a physically grounded comparative framework for AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media. Within the tested assumptions, PINN is the strongest physics-informed baseline and shows the best agreement with the analytical travel-time reference. Transformer-style prediction is the fastest route and is useful for rapid query-based inference. MeshGraphNet remains a useful mesh-aware surrogate candidate, but needs further time-conditioning. FNO should be treated as a diagnostic branch until its output normalization is corrected. Overall, the work demonstrates a careful prototype for comparing neural surrogate approaches rather than a final geophysical simulator.

**Key points:**

- The main contribution is a comparative AI-assisted framework.
- PINN performs best against the travel-time reference.
- Other models remain useful candidates with specific limitations.

**Transition to next slide:**

Finally, I will outline the most important next steps.

---

## Slide 31. Future Work

**Purpose:**

Show how the prototype can be developed after the diploma.

**What to say:**

The first direction is to retrain and recalibrate the two-dimensional FNO route, because its current output scale is not reliable. The second direction is to add stronger time-conditioned rollout behavior to MeshGraphNet. The third direction is to extend the material catalogue with more complete thermoelastic parameters and validation data. It would also be useful to compare the prototype against additional analytical and numerical baselines. The final message is that the prototype estimates directional characteristics of thermoelastic wave propagation within simplified assumptions and provides a basis for further research.

**Key points:**

- Recalibrate FNO and improve time behavior in MeshGraphNet.
- Extend material data and validation sources.
- Add stronger analytical or numerical baselines.



**Transition to next slide:**

This completes the presentation, and I am ready for questions.

---

## Slide 32. Thank You

**Purpose:**

Close the presentation and invite questions.

**What to say:**

Thank you for your attention. I will be happy to answer your questions about the physical assumptions, the model comparison, the dataset, or the interpretation of the results.

**Key points:**

- Close politely.
- Signal readiness to discuss assumptions and limitations.
- Keep answers consistent with the research prototype framing.

**Transition to next slide:**

Questions from the committee.

---

## Possible questions from the committee

### 1. Why did you choose directional prediction instead of full-field simulation?

Because the thesis focuses on a narrower and more defensible research task. Full-field thermoelastic simulation is computationally and physically more complex. Directional prediction allows us to compare source–probe responses, travel time, and model behavior under controlled assumptions.

### 2. Is the prototype validated on real field data?

No. The current work uses COMSOL-derived synthetic data and analytical sanity checks. The results should be interpreted within simplified modelling assumptions. Field data would be an important next step, but it is outside the current scope.

### 3. Why is PINN called a physics-informed baseline?

PINN is called a physics-informed baseline because its training objective includes residual terms connected with thermoelastic relations. This means that physical structure is included in the learning objective, but the result is still interpreted as a baseline model inside the simplified prototype.

### 4. Why does FNO perform poorly in the current results?

The current two-dimensional FNO route has a calibration and output-scale issue. Therefore, it is kept in the comparison as a diagnostic branch. Its architecture is still relevant, but this implementation needs retraining and normalization correction.

### 5. What is the role of the analytical reference $t = r/V_P$ ?

It is a simple travel-time sanity check based on distance and P-wave velocity. It is not a complete thermoelastic validation, but it helps identify whether predicted travel times are physically plausible in a basic isotropic approximation.

### 6. What are the main limitations of the work?

The main limitations are the simplified two-dimensional setting, effective isotropic material parameters, limited synthetic data, fixed-horizon behavior in some surrogate routes, and the absence of direct calibration against real geological measurements.

### 7. What is the main contribution of the diploma work?

The main contribution is a physically grounded comparative framework for AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media. It combines thermoelastic theory, geological material parameters, a controlled dataset, and several neural surrogate routes under one consistent comparison workflow.

## Short version of the speech

Good morning. The topic of our diploma work is *AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media*. The work presents a research prototype for AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media. It is not a site-calibrated simulator. The goal is to estimate selected directional characteristics of thermoelastic wave propagation under simplified assumptions and to compare several neural surrogate approaches.

The physical motivation is that temperature changes in rocks can create thermal expansion, stress, displacement, and wave-like mechanical response. This is relevant for geothermal systems, underground engineering, and geophysical interpretation. A full coupled numerical simulation can be expensive when many materials and source–probe directions must be compared. Therefore, the thesis formulates a narrower prediction task.

The model uses effective geological material parameters such as density, elastic moduli, thermal conductivity, heat capacity, thermal expansion, and P-wave and S-wave velocities. These parameters describe how different rock types respond mechanically and thermally. The simplified formulation is two-dimensional, uses small deformation and linear isotropic thermoelasticity, and does not explicitly resolve pores, fluids, fractures, plasticity, or full anisotropy.

Directional prediction is defined by a source point and a probe point. Their difference gives the direction, distance, and azimuth. The prototype estimates temperature, displacement, travel time, and directional response for that query. The data are based on a controlled COMSOL-derived two-dimensional thermoelastic setup, exported and normalized for several model families.

The comparison includes the Physics-Informed Neural Network, Fourier Neural Operator, MeshGraphNet, and a Transformer-style route. PINN is used as the physics-informed baseline because its objective includes thermoelastic residual terms. FNO, MeshGraphNet, and Transformer-style models are treated as surrogate routes for comparison, not as physically equivalent numerical solvers.

The calculations use forty scenarios per material, four materials, and four model routes, giving six hundred forty model responses. In the travel-time comparison, model-family differences are larger than the material difference in the current prototype. Against the simple analytical reference  $t = r/V_P$ , PINN shows the best agreement and remains within the ten percent band in the balanced sweep. Transformer-style prediction is the fastest route, while MeshGraphNet and FNO need additional development for time behavior and calibration.

The main limitations are the limited synthetic dataset, the simplified two-dimensional assumptions, the current FNO normalization issue, and the fact that the analytical reference is only a sanity check. The conclusion is that the work provides a physically grounded comparative framework for AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media within simplified assumptions. Future work should recalibrate FNO, improve time-conditioned MeshGraphNet behavior, extend material data, and compare against additional analytical or numerical baselines.